

Forward Coverage and Contracting Returns Modulo Nine

Collatz proof project

September 5, 2026

Abstract

We formalize Corollary 5.8 of Monks et al.: every positive natural Collatz orbit visits two modulo nine. Applying the result to every tail gives arbitrarily late visits and a conditional reduction of convergence to this progression. We also bound contracting first returns sharply by sixteen steps and verify their strict descent for seeds above two. This is a Lean formalization of a known theorem; it does not exclude divergence or nontrivial cycles.

Write $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n .

Theorem 1. *For every $n > 0$ and $K \geq 0$, some $t \geq K$ satisfies $T^t(n) \equiv 2 \pmod{9}$.*

Proof. Our existing formal arithmetic results give a tail outside multiples of three and an even value on every natural orbit. On the six remaining residues the possible transitions, with parity suppressed, are

r	1	2	4	5	7	8
$T(r)$	5, 2	1, 8	2	7, 8	8, 2	4, 8.

Assume a tail avoids residue two. The finite transition constraints force its value after three steps to have residue eight: residue four must be excluded since its successor is two, and the remaining non-eight chain is 1, 5, 7, 8. Applying this argument to every further tail shows that all values from time three onward have residue eight. An even value of residue eight has successor residue four, a contradiction. This proves a visit exists. Starting the same argument at $T^K(n) > 0$ gives the stated time bound. $\square$

The even-value lemma uses finite parity injectivity: if a natural seed n had only odd steps, its parity prefix of length $k = n + 2$ would agree with $2^k - 1$, forcing equality of their residues modulo 2^k . Both lie below 2^k , while $n < 2^k - 1$, a contradiction.

Contracting first returns

Suppose $n \equiv 2 \pmod{9}$ and no $T^s(n)$ with $0 < s < t$ has that residue. Let $j = j_t(n)$ count the odd steps and let $e = t - j$ count the even steps. Assign heights to the non-two residues by

$$H(1) = 5, \quad H(5) = 4, \quad H(7) = 3, \quad H(8) = 2, \quad H(4) = 1, \quad H(2) = 0.$$

Away from residue two, an even step decreases height by at least one and an odd step does not increase it. After the first step the height is at most five. Telescoping gives $e \leq 6$, without bounding the odd run at eight.

Theorem 2. *Under these avoidance hypotheses, if $3^j < 2^t$, then $t \leq 16$. If additionally $n > 2$, then $T^t(n) < n$.*

Proof. For $j \geq 11$, induction gives $3^j > 64 \cdot 2^j \geq 2^t$ since $t \leq j + 6$. Thus contraction forces $j \leq 10$ and $t \leq 16$. For descent, the existing growth bound gives

$$2^t(T^t(n) + 1) \leq 3^j(n + 2^{t-j}).$$

A kernel-checked arithmetic table for $0 \leq j \leq t \leq 16$ proves $3^j(573 + 2^{t-j}) < 574 \cdot 2^t$ whenever $3^j < 2^t$. If $n \geq 573$, these inequalities and contraction exclude $T^t(n) \geq n$. For the remaining seeds, a second kernel-checked table tests the exact avoidance and coefficient hypotheses for $n < 573$ and $t \leq 16$. The cutoff is derived from the global inequality; the theorem has no seed-height cutoff in its hypotheses. $\square$

The sixteen-step bound is sharp: 147440 first returns to the progression at 132860, after ten odd and six even steps. Lean checks the full no-intermediate-visit condition. The seed two must be excepted from strict endpoint descent: 2, 1, 2 is a contracting first return. Growing returns also occur, for example 11, 17, 26, 13, 20. No statement here asserts that a given orbit eventually has a contracting first return or that a succession of growing returns is impossible.

Multiple returns: a failed extension and an open test

The avoidance hypothesis cannot be discarded. Lean checks

$$T^{46}(470) = 479, \quad j_{46}(470) = 29, \quad 3^{29} < 2^{46},$$

with both endpoints two modulo nine. An earlier visit is $T^2(470) = 353 < 470$, so this does not refute descent at the *first* cumulative coefficient contraction observed in the progression.

We tested that latter condition for the 999,999 seeds 11, 20, $\dots$, 8999993, with a cap of 3000 steps per seed. Every tested seed reached a sampled contraction within the cap, and every such endpoint was smaller than its starting seed. The longest was $T^{225}(1689023) = 196274$, with 140 odd steps and 45 positive-time visits. Lean separately checks that individual endpoint, its contracting coefficient, and noncontraction at every earlier residue-two visit. The million-scale scan itself is Python evidence, not a Lean theorem. The reproducible program and output are `analysis/residue_return_stopping.py` and its `_results.json` companion. Descent at the first sampled contraction remains an unproved candidate statement. Its eventual existence also remains unproved for arbitrary seeds.

Convergence reduction and remaining gap. If every positive $n \equiv 2 \pmod{9}$ reaches one, then every positive seed reaches one by composition with the hitting time above. That premise is unproved. In particular, the result cannot be combined directly with the least-noncontracting-seed restriction: the property $2^t \leq 3^{j_t(n)}$ for all t is not supplied at the selected residue visit. Coverage and the existence of noncontracting tails are separate existential statements; their witnesses have not been shown to coincide.

Formal artifacts. `Collatz.ResidueCoverage` proves `exists_two_mod_nine`, `arbitrarily_late_two_mod_nine`, and `reaches_one_of_two_mod_nine`. The existing `exists_even_value` is now public. The same module proves the six-even-step bound, the sharp sixteen-step bound, and `contracting_two_mod_nine_segment_descends`. The proofs use exact arithmetic and existing parity results, with no imported literature axiom or finite seed cutoff.

References

- [1] K. Monks, K. G. Monks, K. M. Monks, and M. Monks, *Strongly sufficient sets and the distribution of arithmetic sequences in the $3x + 1$ graph*, arXiv:1204.3904v2 (2012), Corollary 5.8.
<https://arxiv.org/abs/1204.3904>.